

§1 Warm up

Taylor: $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$ $\frac{1}{1-x} = 1 + x + x^2 + x^3 + o(x^3)$

$\sin x = x - \frac{x^3}{6} + o(x^3)$

$\cos x = 1 - \frac{x^2}{2} + o(x^2)$

$\frac{1}{1+x} = 1 - x + x^2 + o(x^2)$

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3)$

$y' + p(x)y = q(x)$

$y = e^{-\int p(x)dx} (\int q(x) e^{\int p(x)dx} dx + C)$

$\vec{f} = \frac{-i\hbar}{2m} (\nabla^2 \psi - \psi \frac{\partial^2}{\partial x^2})$

§2

A1

$\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^3} = \lim_{x \rightarrow 0} \frac{x - \frac{x^3}{6} - x(1 - \frac{1}{2}x^2) + o(x^3)}{x^3} = \frac{1}{3}$

A2

$\det A = 2 \cdot (-1)^{2+2} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 6$

$\det(A - I) = (2-1)(-1)^{2+2} \begin{vmatrix} 2-1 & 1 \\ 1 & 2-1 \end{vmatrix} = 0$

$\lambda_1 = 1$
 $\lambda_2 = 2$
 $\lambda_3 = 3$

$\lambda_1 \lambda_2 = 6 = \det A$

$\lambda_1 + \lambda_2 + \lambda_3 = 6 = \text{tr} A$

A3

$\frac{dy}{dx} (2xy = 1)$

$y = e^{-\int 2x dx} (\int 1 \cdot e^{\int 2x dx} dx + C)$

$\psi(0) = 0 \Rightarrow C = 0$
 $y = e^{x^2} \int e^{-x^2} dx$

$y = e^{x^2} \int_0^x e^{-t^2} dt$

B1

错误 error

因为 Heisenberg 统计是 $(\psi|0)$ 体系不变 $A_n = e^{iHt/n} A e^{-iHt/n}$ 随时间变化。

B2

$\nabla \cdot \vec{j} + \frac{\partial \rho}{\partial t} = 0$

$\rho = |\psi|^2$ 为概率密度

$\frac{\partial \rho}{\partial t} = -\nabla \cdot \vec{j}$

概率密度 (取相同标度) 为概率流密度

物理定律是全空间 概率守恒。

Ehrenfest ... 也有类似。

一实验发现粒子波函数的概率分布。

波函数流出与界的通量密度。

§3

C1 (V)

$$\frac{d\hat{p}_H}{dt} = \frac{1}{i\hbar} [\hat{x}_H, \hat{p}] = \frac{1}{i\hbar} [\hat{x}_H, \hat{p}] = \frac{1}{i\hbar} \{ \hat{p}, \hat{x}_H \}$$

$$= \frac{1}{i\hbar} \{ \hat{p}, \hat{x}_H \} = \frac{1}{i\hbar} \{ \hat{p}, \hat{x}_H \} = \frac{1}{i\hbar} \{ \hat{p}, \hat{x}_H \}$$

$$= \frac{\hat{p}_H}{m}$$

$$(4) \frac{d\hat{p}_H}{dt} = \frac{m\omega^2}{2i\hbar} [\hat{x}_H^2, \hat{x}_H] + [\hat{p}_H, \hat{x}_H] \hat{x}_H = -m\omega^2 \hat{x}_H$$

(IV)

$$\frac{d\hat{p}_H}{dt} = \frac{m\omega^2}{2i\hbar} [\hat{x}_H^2, \hat{x}_H] + [\hat{p}_H, \hat{x}_H] \hat{x}_H = -m\omega^2 \hat{x}_H$$

Zustand

§D

$$\frac{d\hat{x}_H}{dt} = \frac{1}{m} \frac{d\hat{p}_H}{dt} = \frac{1}{m} \cdot (-m\omega^2 \hat{x}_H) = -\omega^2 \hat{x}_H$$

$$\frac{d\hat{x}_H}{dt} + \omega^2 \hat{x}_H = 0$$

$$I = \int_{-\infty}^{+\infty} e^{-ax^2} dx \quad (a > 0)$$

$$I^2 = \left(\int_{-\infty}^{+\infty} e^{-ax^2} dx \right) \left(\int_{-\infty}^{+\infty} e^{-ay^2} dy \right) = \iint_{\mathbb{R}^2} e^{-a(x^2+y^2)} dx dy$$

$$= \iint_{\mathbb{R}^2} e^{-ar^2} r dr d\theta = 2\pi \int_0^{+\infty} e^{-ar^2} r dr$$

$$= \pi \int_0^{+\infty} e^{-ar^2} dr$$

$$= \pi \left(-\frac{1}{2a} \right) e^{-ar^2} \Big|_0^{+\infty}$$

$$= \frac{\pi}{2a}$$

$$I = \frac{\pi}{2a}$$

$$I = \sqrt{\frac{\pi}{a}}$$